

Nutritional risk factors for the development of chronic obstructive pulmonary disease (COPD) in male smokers.

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BACKGROUND & AIMS: The aim of this study was to investigate whether nutritional risk factors, especially black tea consumptions, are inversely associated with the development of chronic obstructive pulmonary disease (COPD) in male smokers. METHODS: Forty male smokers with clinical diagnosis of COPD (Group-I (GI)) and 36 healthy smokers without COPD (Group-II (GII)) were included in this study. We compared the dietary habits and food intakes of the two groups using an adaptation of the Arizona Food Frequency Questionnaire (AFFQ). Question form included a list of 65 food items formed from five main food groups (grain, meat and alternatives, dairy products, vegetables-fruits and fat) and 25 dietary habits. The data were evaluated by binary logistic regression analysis, receiver operating characteristic (ROC) curve, Kolmogorov-Smirnov, Student's t, Mann-Whitney, and Chi-square tests. RESULTS: When both groups compared, black tea consumptions (GI-700ml; GII-1600ml (OR: 0.635, $P<0.001$)), vegetable fruits scores (GI-54.30; GII-63.81 (OR: 0.863, $P<0.001$)), regularly breakfast habit (GI-24 patients; GII-36 cases (OR: 0.549, $P<0.001$)) and eating salty (GI-22 patients; GII-5 cases ($P<0.001$)) made significant differences. In ROC curves, the area under the curve of black tea (0.898 (95% CI: 0.819-0.977) and vegetables-fruits (0.833 (95% CI: 0.727-0.938) provided high accuracy to distinguish between COPD group and controls ($P<0.001$). CONCLUSIONS: High intakes of black tea and vegetables-fruits consumptions may be protecting male smokers from developing COPD.